

HIGH RESPONSE RADIOTHERAPY IN PATIENTS RECEIVING ALPHA-TOCOPHEROL DURING TREATMENT IN LUNG CANCER

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ABSTRACT

Fifty six patients, with an established diagnosis of lung cancer were randomized into two groups. They received radiotherapy with and without alpha-tocopherol. 5.26% relative response, 0.00% complete response, 5.26% no response and 89.77% progressive disease were noted in patients who received placebo during radiotherapy, while 29.62% relative response, 7.40% complete response, 7.40% no response and 62.96% progressive disease were found in patients who received alpha-tocopherol during this therapy. Similarly no recurrence and 26.30% metastasis (in brain, bone, liver and lymph node after an average duration of 39 days) with poor survival was noted in the former group, while 3.70% recurrence and 14.81% metastasis (in brain and bone after an average duration of 60 days) with better survival was found in later group.

INTRODUCTION

The total outcome of all the therapeutic efforts for the treatment of lung cancers is still not hopeful. It accounts 35% of all male cancer fatalities and 15% of all female cancer deaths (Cancer Research Campaign, 1987). Surgery is in the first line of management. The majority of reports on 30 days, operative mortality rate for the resection of lung cancer, based on surgical series showed 2.1% to 12.4% mortality over the past 10 to 25 years respectively (Fryjordent and Klevmark, 1971; Weiss, 1974; Nagasaki et al., 1982). Radiotherapy is also an important modality of treatment in lung cancers. Unfortunately the outlook of radiotherapy is also poor. The Radiation Therapy Oncology Group (RTOG) randomized 551 patients with unresectable or inoperable new small cell cancer to different types of potentially curative treatment with radioactive therapy. The intrathoracic failure rate within the irradiated volume was found to be varied in a dose related manner. 48% for 4000c Gy continuous, 38% for 5000c Gy continuous (or 4000c Gy split course) and 27% for 6000c Gy continuous. The failure rate in the non-irradiated lung was 25% to 38% irrespective to radiation dose. 75% to 80% of all the patients developed distant metastasis. 16% of those with squamous cell carcinoma and 30% of those with adenocarcinoma and 54% of large cell carcinoma had brain metastasis (Perez et al., 1987). Similarly the results of chemotherapy in such cases are also discouraging. In addition to disappointing results (Bakowski and Cronch, 1983; Rapp et al., 1988), not all the patients suit for chemotherapy because of general medical problems

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and poor performance status (Thatcher, 1990). In developing countries like Pakistan, radiotherapy is the most important modality, as surgery is not possible due to diagnosis at advanced stage and poor facilities and chemotherapy is not feasible due to poor performance status of these patients. Even dose of radiation has to be compromised due to poor tolerability of these patients. Therefore an attempt has been made to enhance the outcome of radiotherapy, in our study.

MATERIALS AND METHODS

From January 1991, upto January 1992, fifty six patients, who had an established diagnosis of lung cancer were recruited in this study. The ages ranged from 35 to 78 years with a mean of 57.29 years. They were candidates for Radiotherapy for their primary lesions and came to Institute of Radiotherapy, Jinnah Postgraduate Medical Centre, Karachi, after being referred from different Medical and Surgical units. They were in stage II and III. They were medically fit and without any contraindication for radiation treatment. Karnofsky scale (Karnofsky and Burchenal, 1949) was applied to grade their physical status. MRC, respiratory score (British Journal of Cancer, 1979) was determined to mark their respiratory status. The extents of their lesions, were determined by clinical examination, roentgenography, tomography or isotope scanning. Responses were assessed using UICC criteria (UICC, 1987) and toxicities were determined by WHO guidelines (Miller et al., 1981) guidelines. The important informations about their disease were obtained from their hospital records.

They were randomized into two groups, receiving same dose of maroon coloured capsules having identical size, weight and consistency. The capsules contained either 200 mg alpha- tocopherol or placebo. Each patient received one capsule TDS. They received this dose for two months, during which they had radiation treatment.

A telecobalt unit was used for radiotherapy. Each patient received a course of 45-50 Gy in five days a week and treatment was completed in four to five weeks. The field of radiation was adjusted according to roentgenographic appearance of the lesion. Each patient received supportive treatment, during the therapy, if required.

Assessment of response and the toxicity was carried out on weekly basis.

RESULTS

Table I shows characteristics of the patients.

Table II shows the response of radiotherapy in two groups of the patients. A high response i.e. 29.62% 'relative response' and 7.4% 'complete response' were noted in patients receiving alpha- tocopherol, rather than 5.2% 'relative response' and 00% 'complete response', which were found in patients receiving placebo. Similarly 5.26% 'no response' and 87.77% 'progressive disease' were noted in patients receiving alpha-tocopherol com-

pared to 7.40% 'no response' and 62.6% 'progressive disease', were found in patients receiving placebo.

Table III represents the recurrence or metastasis, during one year follow up of the patients. No recurrence was noted in patients receiving placebo, as no patient had complete response in this group. But 3.70% recurrence was noted in patients receiving alpha-tocopherol. 26.30% patients had metastasis in the group in which the patients received placebo, while, 14.81% patients had metastasis in the group, in which the patients received alpha-tocopherol. A comparison of the two groups shows less metastasis in the patients who were treated with alpha- tocopherol during radiation.

Table IV shows sites of metastasis. The patients who received placebo during treatment, had 10.52% metastasis to brain, 5.26% to bones, 5.26% to liver and 5.26% to lymph glands, while the patients who received alpha-tocopherol had 7.40% metastasis to brain, 7.40% to bone, 00% to liver and 00% to lymph nodes.

Table V shows duration of distant metastasis. An average 2.5 months elapsed for the distant metastasis to appear in the patients receiving placebo, while an average 3.5 months elapsed for the distant metastasis to appear in the patients receiving alpha-tocopherol.

Table VI represents that survival of the patients who received placebo was poor, as only 10.52% patients survived for more than 6 months, 36.84% for more than 3 but less than 6 months and 52.16% for less than 3 months. The survival of the patients who received alpha-tocopherol was relatively better, as 25.92% patients survived for more than 6 months, 25.92% for more than 3 but less than 6 months, and 48.14% for less than 3 months.

Table VII represents the overall survival of the patients of both the groups. 16.40% patients survived for more than 6 months, 41.30% for more than 3 months but less than 6 months, and 41.30% for less than 3 months.

Table VII reveals the radiations induced toxicities, which were almost same in both the groups of the patients.

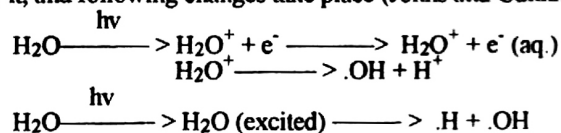
DISCUSSION

More than 50-fold increase in incidence of lung cancers is found on statistical analysis of 25 years data (Thatcher, 1990). The cancer treatment schemes, based on a selection of the most recent literature, still focus on radiotherapy as an important tool of treatment. This study shows a significant improvement in the response to radiation, in patients who received additional doses of alpha-tocopherol. This conclusion is not altogether a surprising result of this clinical trial, as the following two vitro experiments support this finding strongly:

- (I) An experiment was performed by Kagerud and his colleagues. In rats, the tumours were implanted in one hind leg. MDO electrode was used to measure tissue oxygen pressure. A significant tissue hypoxia developed under tourniquet of the tumour bearing leg of the animals. Pretreatment with tocopherol did not influence the tumour oxygen pressure. The tumours were treated by radiotherapy, less growth retardation was noted in tumours which were not pretreated by alpha-tocopherol, while significant growth retardation was seen in tumours which were pretreated by alpha-tocopherol (Kagerud et al., 1981).
- (II) Another experiment was done by Kagerud and Peterson. In this experiment, the influence of alpha-tocopherol, administered intramuscularly in different single doses, on the effect of local irradiation of a transplanted sarcoma, was investigated. Alpha-tocopherol enhanced significantly the tumour growth retardation induced by irradiation (Kagerud and Peterson, 1981).

Similarly, the interactions of alpha-tocopherol at cellular level lead to the idea that this compound makes the malignant cells, more susceptible to be destroyed by radiation, by the following facts based on a number of direct or indirect inter-related studies:

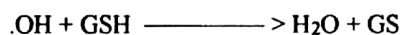
- (I) The relationship of alpha-tocopherol and members of the co-enzyme 'Q' group is established. The structure of Q is very similar to this compound (Mayes, 1990). Abnormal function of co-enzyme 'q' in electron transport or in oxidative phosphorylation might account for different symptoms seen in alpha-tocopherol deficiency (Machlin, 1980). Similarly the members of this co-enzyme group, cure or protect against this deficiency in several animal species (Bieri and Farrell, 1976). The compound hexahydro co-enzyme 'Q', which is very similar chemically to a naturally occurring co-enzyme 'Q' has proved to cure nutritional muscular dystrophy induced by alpha-tocopherol deficiency in rabbits (West et al., 1989). The co-enzyme 'Q' (Ubiquinone) is a component of the terminal respiratory chain (West et al., 1989). It means that alpha-tocopherol also plays an important part in respiratory process. It has been proved that tissue oxygenation improves the radiation response (Mattram, 1936; Read, 1952; Gray et al., 1953). Therefore it can be expected that alpha-tocopherol can enhance effect of radiation by improving respiratory process of malignant cells. The ability of this compound to improve tissue oxygenation has also proved by measurement of oxygen pressure in the tumour pretreated by alpha-tocopherol before inducing experimental hypoxia (Kagerud et al., 1981).
- (II) Radiation process produces a number of physical and chemical events, in the tissues, in rapid succession. Since cells are more than 70% water, so most of the energy is absorbed by it, and following changes take place (Johns and Cunningham, 1983).



The net result is the activation of atoms or molecules (free radicals) which are extremely reactive and interact with organic molecules and macromolecules (Baufi, 1988) leading to biological damage. It is generally believed that $\cdot\text{OH}$ is the most effective causing biological damage (Johns and Cunningham, 1983). This radical tends to be reduced to form OH^- which is, inactive biologically,



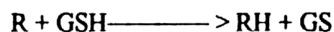
This fact is also supported by considering the mechanism of radiation protection by intracellular glutathione (GSH); which reduces the active $\cdot\text{OH}$ to produce water molecule (Wassermann, 1990).



High alpha-tocopherol concentration, produces antioxidant environment, and so save the products of radiated water.

Two other possibilities are required to be emphasized at this stage. First, the alpha-tocopherol does not scavenge the activated $\cdot\text{OH}$, because the antioxidant nature of this compound is based upon its OH group (West et al., 1989) and energetically no chemical bonding is possible between these two species. Second the alpha-tocopherol saves the products of radiated water, by inhibiting its reduction due to creating an antioxidant tissue environment. The reduction of the activated molecules takes place by electrons which are given by free radicals. The alpha-tocopherol molecules or their products bond probably covalently to these free radicals, so make them unable to donate electrons. A fact, due to its antioxidant action, which has already been proved (Fritsma, 1983). This mechanisms is a similar mechanism, as we expected in the role of alpha-tocopherol in reducing carbon tetrachloride induced toxicities, which have been shown in the experimental work of this study.

Another benefit of high alpha-tocopherol levels, during the radiation is the stabilization of the damage of the cellular components. The $\cdot\text{OH}$ extracts a hydrogen atom from DNA, leaving a radical site on it (Johns and Cunningham, 1983). This damage can be repaired by providing hydrogen to the damaged tissue component (Wassermann, 1990).



Alpha-tocopherol can probably stabilizes the damage by providing OH^- , on which its most of properties are based (West et al., 1989).

- (III) An elevated concentration of skeletal muscles deoxyribonucleic acid (DNA) and an increase in DNA and ribonucleic acid (RNA) in bone marrow as well as an increase bone marrow RNA/DNA ratio was found in alpha-tocopherol deficient monkeys (West

et al., 1989). Perhaps the repair of DNA and RNA damage induced by radiations is blocked in an excess of alpha-tocopherol during radiation. This is important to mention that tumour response is very greatly influenced by repair mechanism and radiation resistant tumours have strong repair mechanism (Welchselbaum et al., 1977).

A better response in terms of metastasis and survival is perhaps, if a sequel of better immune response which is developed in patients having high levels of alpha-tocopherol than those who have low levels of this compound. The importance of immune reaction against malignant cells is evident by following facts:

- (I) Ioachim (1976) presented histologic examination of certain tumours in man, which had a heavy mononuclear infiltrate (lymphocytes, macrophages, plasma cells) in and about the tumour, particularly in certain carcinomas of the lung, testis and breast, malignant melanomas, and Hodgkin's disease.
- (II) Everson and Cole (1966) reported 130 instances of spontaneous regression of metastatic malignant tumours, half of them included choriocarcinomas, hypernephromas, neuroblastomas, and malignant melanomas, for which immune responses were suspected.
- (III) Burnet (1970) brought forward the concept of immunologic surveillance against tumours, which is based upon these reactions. Although this concept had to face challenges, but it was firmly supported by numerous evidences (Hersh, 1975).

Similarly importance of alpha-tocopherol in development of an immune reaction, is supported by:

- (I) Deshazo and colleagues (1981) showed the involvement of monocyte derived toxic oxygen metabolites in lymphocyte dysfunction of Hodgkin's disease. They also showed augmentation of lymphocyte proliferation with the addition of catalase or alpha-tocopherol in the culture system.
- (II) Combined administration of vitamin E and C resulted in significantly better clinical improvement in patients with cholasma and pigmented contact dermatitis, an allergic manifestation (Hayakawa et al., 1981). Similarly vitamin E is found to improve immune system of patients (Haberal et al., 1988).
- (III) In Acquired Immune Deficiency Syndrome (AIDS), the cell membrane distortion of the T-4 cell protein receptor for genetically related macrophages factor (GRF) is one of the possible effects of HIV virus on T-cells. The integrity of this specific T-cell receptor takes on profound significance due to its relationship to soluble antigen. Alpha-tocopherol has been shown to assist in the maintenance of cell membrane structure (Hollin, 1988).

(IV) Vitamin E supplementation is found to suppress prostaglandin E₁(2) synthesis to enhance immune response of aged mice (Meydani et al., 1986). Similarly this compound prevents a decline of natural killer cells activity under conditions where free radical formation by activated macrophages and polymorphonuclear leucocytes may be enhanced (Meydani et al., 1988).

Table I
Patients characteristics

Total number of patients entered	= 56
No. of patients expired	= 07
No. of patients dropped due to non-compliance	= 03
Evaluable patients	= 46
Age	= 57.29 years (Range 35-78 years)
Sex (M/F)	= II and III
Performance status	= 40-90% (on Karnofsky scale)
Respiratory status	
35.7% grade I	
23.8% grade IV	
19.0% grade III	
11.9% grade V	
09.5% grade II (MRC-Score)	
Histopathology of lesion	=
Squamous cell carcinoma	= 31 patients
Large cell carcinoma	= 06 patients
Small cell carcinoma	= 06 patients
Adenocarcinoma	= 01 patients
Metastasis in gland or malignant cells in pleural effusion	= 02 patients
Main symptoms	=
Pain in chest	= 46.0% patients
Hemoptysis	= 30.0% patients
Breathlessness	= 11.0% patients
Cough	= 06.06% patients
(productive or non-productive)	
Hoarseness	= 04.0% patients
Swelling or mass	= 04.0% patients

(Most of patients presented with more than one symptoms)

Table II
Response of radiotherapy in 46 evaluable patients

Group	No. of Patients	CR+PR	CR	NR	PD
Placebo	19	01 (05.26%)*	00 (00%)	01 (05.26%)	17 (89.77%)
Alpha-tocopherol	27	08 (29.62%)	02 (07.40%)	02 (07.40%)	15 (62.96%)

*No. of patients (Percentage)

CR = Complete remission, PR = Partial response, NR = No response, PD = Progressive disease.

Table III
Recurrence or metastasis in 46 evaluable patients

Group	No. of patients	Recurrence	Metastasis
Placebo	19	00 (00.00%)	05 (26.30%)
Alpha-tocopherol	27	01 (03.70%)	04 (14.81%)

Table IV
Site of metastasis

Group	Site of Metastasis				
	Brain	Bone	Liver	Lymph glands	Others
Placebo	02 (10.52%)	01 (05.26%)	01 (05.26%)	01 (05.26%)	00 (00.00%)
Alpha-tocopherol	02 (07.40%)	02 (07.40%)	00 (00.00%)	00 (00.00%)	00 (00.00%)

